Chpater3 Electricity & DC Circuits

Lesson2 Electric Current, Resistance & DC Circuits

1. If a current of $80.0\ \text{mA}$ exists in a metal wire, **(a)** how many electrons flow past a given cross section of the wire in $10.0\ \text{min}$? **(b)** In what direction do the electrons travel with respect to the current?

2. A copper wire has a circular cross section with a radius of $1.25\ \text{mm}$. **(a)** If the wire carries a current of 3.70 A, find the drift speed of the electrons in the wire. (See **Example 17.2** for relevant data on copper.) **(b)** All other things being equal, what happens to the drift speed in wires made of metal having a larger number of conduction electrons per atom than copper? Explain.

3. The current supplied by a battery in a portable device is typically 0.15 A. Find the number of electrons passing through the device in one hour.

4. A wire of diameter $0.800\ \text{mm}$ and length $25.0\ \text{m}$ has a measured resistance of 1.60 V. What is the resistivity of the wire?

5. A potential difference of 12 V is found to produce a current of 0.40 A in a 3.2-m length of wire with a uniform radius of $0.40\ \text{cm}$. What is **(a)** the resistance of the wire? **(b)** The resistivity of the wire?

6. Suppose your waffle iron is rated at $1.00\ \text{kW}$ when connected to a 1.20×10^2 V source. **(a)** What current does the waffle iron carry? **(b)** What is its resistance?

7. If electrical energy costs $\$0.12$ per kilowatt-hour, how much does it cost to **(a)** burn a 100-W lightbulb for 24 h? **(b)** Operate an electric oven for 5.0 h if it carries a current of 20.0 A at 220 V?

8. The potential difference across a resting neuron in the human body is about $75.0\ \text{mV}$ and carries a current of about $0.200\ \text{mA}$. How much power does the neuron release?

9. If a battery is rated at 60.0 A? h, how much total charge can it deliver before it goes "dead"?

10. A car owner forgets to turn off the headlights of his car while it is parked in his garage. If the 12.0-V battery in his car is rated at 90.0 A? h and each headlight requires 36.0 W of power, how long will it take the battery to completely discharge?

11. Four resistors are arranged as shown in **Figure 18.5a**. Find **(a)** the equivalent resistance of the circuit and **(b)** the current in the circuit if the closed-circuit terminal voltage of the battery is 6.0 V. **(c)** Calculate the electric potential at point A if the potential at the positive terminal is 6.0 V. **(d)** Suppose the open circuit voltage, or emf e, is 6.2 V. Calculate the battery's internal resistance.

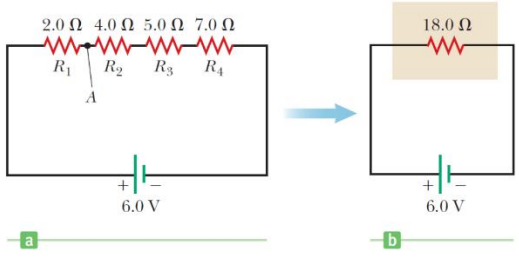


Figure 18.5 (Example 18.1) (a) Four resistors connected in series. (b) The equivalent resistance of the circuit in (a).

12. A circuit consists of a battery with a closed circuit, a terminal voltage of 12.0 V, and 3.0-V, 6.0-V, 8.0-V, and 9.0-V resistors connected in series, oriented as in **Figure 18.5a**, with the battery in the bottom of the loop, positive terminal on the left, and resistors in increasing order, left to right, in the top of the loop. Calculate **(a)** the equivalent resistance of the circuit, **(b)** the current, and **(c)** the total power dissipated by the load resistors. **(d)** What is the electric potential at a point between the 6.0-V and 8.0-V resistors, if the electric potential at the positive terminal is 12.0 V? **(e)** If the battery has an emf of 12.1 V, find the battery's internal resistance.

13. Three resistors are connected in parallel as in **Figure 18.8**. A potential difference of 18 V is maintained between points a and b. **(a)** Find the current in each resistor. **(b)** Calculate the power delivered to each resistor and the total power. **(c)** Find the equivalent resistance of the circuit. **(d)** Find the total power delivered to the equivalent resistance.

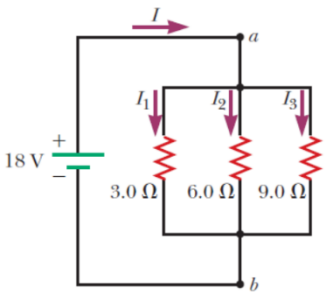


Figure 18.8 (Example 18.2) Three resistors connected in parallel. The voltage across each resistor is 18 V.

(*Continued*)

14. Suppose the resistances in the example are 1.0 V, 2.0 V, and 3.0 V, respectively, and a new voltage source is provided. If the current measured in the 3.0-V resistor is 2.0 A, find **(a)** the potential difference provided by the new battery and the currents in each of the remaining resistors, **(b)** the power delivered to each resistor and the total power, **(c)** the equivalent resistance, and **(d)** the total current and the power dissipated by the equivalent resistor.

15. A battery having an emf of 9.00 V delivers 117 mA when connected to a 72.0-V load. Determine the internal resistance of the battery.

16. **(a)** Find the current in a 8.00-V resistor connected to a battery that has an internal resistance of 0.15 V if the voltage across the battery (the terminal voltage) is 9.00 V. **(b)** What is the emf of the battery?

17. **(a)** Find the equivalent resistance between points a and b in **Figure P18.5**. **(b)** Calculate the current in each resistor if a potential difference of 34.0 V is applied between points a and b.

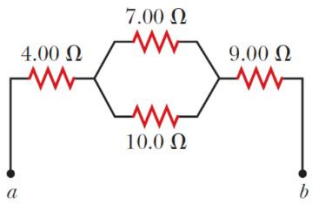


Figure P18.5

18. Consider the combination of resistors shown in **Figure P18.6**. **(a)** Find the equivalent resistance between point a and b. **(b)** If a voltage of 35.0 V is applied between points a and b, find the current in each resistor.

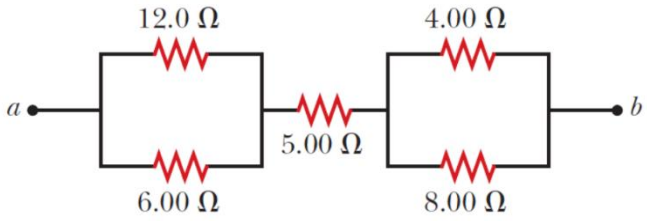


Figure P18.6

19. Two resistors connected in series have an equivalent resistance of 690 V. When they are connected in parallel, their equivalent resistance is 150 V. Find the resistance of each resistor.

20. Consider the circuit shown in **Figure P18.8**. (a) Calculate the equivalent resistance of the 10.0-V and 5.00-V resistors connected in parallel. **(b)** Using the result of part **(a),** calculate the combined resistance of the 10.0-V, 5.00-V, and 4.00-V resistors. **(c)** Calculate the equivalent resistance of the combined resistance found in part **(b)** and the parallel 3.00-V resistor. (d) Combine the equivalent resistance found in part **(c)** with the 2.00-V resistor. **(e)** Calculate the total current in the circuit. **(f)** What is the voltage drop across the 2.00-V resistor? **(g)** Subtracting the result of part **(f)** from the battery voltage, find the voltage across the 3.00-V resistor. **(h)** Calculate the current in the 3.00-V resistor.

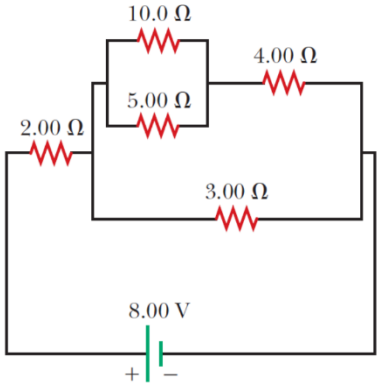


Figure P18.8

21. Consider the circuit shown in **Figure P18.9**. Find **(a)** the potential difference between points a and b and **(b)** the current in the 20.0-V resistor.

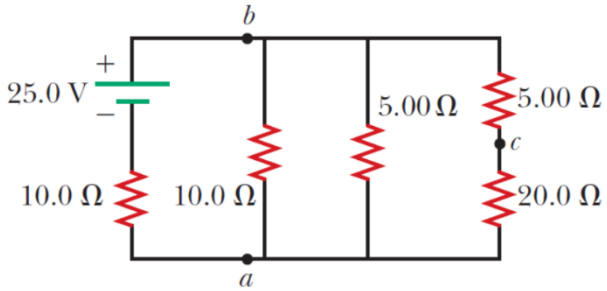


Figure P18.9

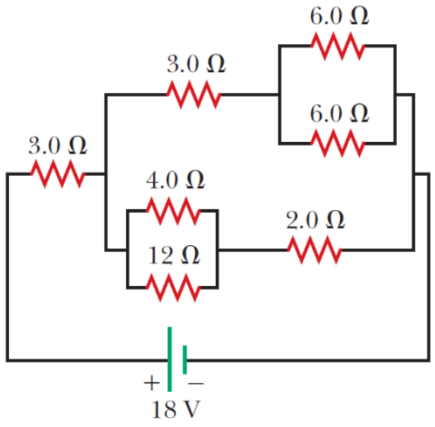


Figure P18.13

22. Find the current in the 12-V resistor in Figure P18.13.

23. A current $I = 15\ \mathrm{A}$ is directed along the positive x-axis and perpendicular to a magnetic field. A magnetic force per unit length of $0.12\ \mathrm{N/m}$ acts on the conductor in the negative y-direction. Calculate the magnitude and direction of the magnetic field in the region through which the current passes.

24. A straight wire carrying a 3.0-A current is placed in a uniform magnetic field of magnitude $0.28\ \mathrm{T}$ directed perpendicular to the wire. **(a)** Find the magnitude of the magnetic force on a section of the wire having a length of $14\ \mathrm{cm}$. **(b)** Explain why you can't determine the direction of the magnetic force from the information given in the problem.

25. A wire carries a current of $10.0\ \mathrm{A}$ in a direction that makes an angle of $30.0°$ with the direction of a magnetic field of strength $0.300\ \mathrm{T}$. Find the magnetic force on a 5.00-m length of the wire.